

## NORTH CAPE, PE, Canada

WMO: 719870

Lat: 47.0581N

Lon: 63.9969W

Elev: 8

StdP: 101.23

Time Zone: -4.00 (NAA)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | -18.2      | -16.0 | -22.2                         | 0.5 | -17.6 | -19.6 | 0.7 | -15.5 | 19.4                  | -7.4 | 18.0 | -7.7 | 8.9                   | 290  | 0.796 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 6.1                    | 26.0            | 21.0 | 24.4 | 20.3 | 22.9 | 19.6 | 22.1                | 24.8 | 21.1 | 23.4 | 20.2 | 22.3 | 5.8                  | 220  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 21.2                            | 15.8 | 23.5 | 20.3 | 15.0 | 22.6 | 19.4 | 14.2 | 21.7 | 65.0          | 25.0 | 61.4 | 23.4 | 58.3 | 22.4 | 27.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 18.0          | 15.9 | 13.9 | -20.7                      | 30.3 | 2.7                | 1.2 | -22.7                                              | 31.2 | -24.3      | 31.9 | -25.8      | 32.6 | -27.8      | 33.5 |    |    |
| (5)               |      |      | -21.0                      | 23.8 | 2.7                | 1.2 | -22.9                                              | 24.6 | -24.5      | 25.3 | -26.0      | 26.0 | -28.0      | 26.9 |    |    |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 6.3   | -6.4  | -6.9  | -3.0  | 2.2   | 8.2   | 14.1  | 19.3  | 19.2  | 15.3  | 9.7   | 3.9   | -1.5 | (6)  |
| (7)  |                                                                                  | DBStd       | 9.83  | 5.16  | 4.85  | 4.01  | 3.28  | 3.74  | 3.65  | 2.65  | 2.50  | 2.95  | 3.17  | 3.75  | 3.83 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 2298  | 509   | 473   | 404   | 234   | 82    | 6     | 0     | 0     | 1     | 43    | 187   | 358  | (8)  |
| (9)  |                                                                                  | HDD18.3     | 4521  | 767   | 706   | 663   | 483   | 313   | 134   | 20    | 19    | 98    | 267   | 434   | 616  | (9)  |
| (10) |                                                                                  | CDD10.0     | 929   | 0     | 0     | 0     | 1     | 27    | 129   | 288   | 285   | 161   | 35    | 3     | 0    | (10) |
| (11) |                                                                                  | CDD18.3     | 111   | 0     | 0     | 0     | 0     | 0     | 7     | 50    | 46    | 8     | 0     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 262   | 0     | 0     | 0     | 0     | 2     | 22    | 129   | 98    | 11    | 0     | 0     | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 31    | 0     | 0     | 0     | 0     | 0     | 2     | 16    | 11    | 1     | 0     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 6.8   | 8.2   | 7.0   | 6.8   | 6.2   | 5.7   | 5.5   | 5.2   | 5.6   | 6.7   | 7.8   | 8.4   | 9.0  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 1282  | 116   | 99    | 103   | 95    | 97    | 96    | 84    | 92    | 111   | 126   | 129   | 135  | (15) |
| (16) |                                                                                  | PrecMax     | 1592  | 182   | 181   | 203   | 178   | 204   | 174   | 162   | 179   | 249   | 237   | 231   | 257  | (16) |
| (17) |                                                                                  | PrecMin     | 1002  | 46    | 30    | 37    | 40    | 43    | 37    | 42    | 34    | 34    | 36    | 69    | 63   | (17) |
| (18) |                                                                                  | PrecStd     | 137   | 30    | 34    | 38    | 29    | 36    | 35    | 30    | 37    | 53    | 51    | 38    | 42   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 8.1   | 6.7   | 8.0   | 16.4  | 22.0  | 25.9  | 28.5  | 28.1  | 24.6  | 20.2  | 15.5  | 10.0 | (19) |
| (20) |                                                                                  |             | MCWB  | 7.8   | 6.0   | 6.3   | 12.8  | 15.2  | 19.9  | 22.1  | 22.3  | 21.0  | 16.1  | 13.8  | 9.4  | (20) |
| (21) |                                                                                  | 2%          | DB    | 4.1   | 3.1   | 5.2   | 11.4  | 18.2  | 23.2  | 26.5  | 25.7  | 22.2  | 16.9  | 12.5  | 6.7  | (21) |
| (22) |                                                                                  |             | MCWB  | 3.8   | 2.4   | 3.9   | 8.8   | 13.9  | 18.6  | 21.2  | 21.1  | 19.2  | 14.6  | 11.5  | 6.2  | (22) |
| (23) |                                                                                  | 5%          | DB    | 1.5   | 1.1   | 3.2   | 8.2   | 15.7  | 21.1  | 24.7  | 24.2  | 20.9  | 15.2  | 10.5  | 4.7  | (23) |
| (24) |                                                                                  |             | MCWB  | 1.0   | 0.7   | 2.2   | 6.3   | 12.5  | 17.6  | 20.6  | 20.4  | 18.4  | 13.5  | 9.6   | 4.2  | (24) |
| (25) |                                                                                  | 10%         | DB    | 0.0   | -0.3  | 1.8   | 6.1   | 13.4  | 19.3  | 23.1  | 22.7  | 19.5  | 14.0  | 8.6   | 3.3  | (25) |
| (26) |                                                                                  |             | MCWB  | -0.4  | -0.8  | 1.0   | 4.6   | 11.1  | 16.6  | 19.9  | 19.7  | 17.4  | 12.5  | 7.5   | 2.6  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 7.9   | 6.3   | 6.6   | 13.1  | 16.9  | 21.8  | 23.4  | 23.5  | 21.7  | 17.5  | 14.3  | 9.4  | (27) |
| (28) |                                                                                  |             | MCDWB | 8.1   | 6.8   | 7.5   | 15.5  | 19.8  | 24.1  | 26.8  | 26.2  | 23.7  | 19.0  | 15.2  | 9.9  | (28) |
| (29) |                                                                                  | 2%          | WB    | 3.9   | 2.6   | 4.1   | 9.2   | 14.8  | 19.7  | 22.2  | 22.0  | 20.1  | 14.9  | 11.6  | 6.3  | (29) |
| (30) |                                                                                  |             | MCDWB | 4.1   | 3.0   | 4.9   | 10.9  | 17.2  | 22.0  | 25.0  | 24.7  | 21.8  | 16.1  | 12.4  | 6.7  | (30) |
| (31) |                                                                                  | 5%          | WB    | 1.1   | 0.8   | 2.3   | 6.5   | 13.0  | 18.2  | 21.3  | 21.0  | 18.8  | 13.8  | 9.7   | 4.3  | (31) |
| (32) |                                                                                  |             | MCDWB | 1.4   | 1.0   | 3.0   | 7.9   | 15.1  | 20.4  | 23.7  | 23.3  | 20.2  | 14.9  | 10.5  | 4.6  | (32) |
| (33) |                                                                                  | 10%         | WB    | -0.4  | -0.7  | 1.0   | 4.6   | 11.3  | 16.8  | 20.4  | 20.2  | 17.7  | 12.7  | 7.8   | 2.7  | (33) |
| (34) |                                                                                  |             | MCDWB | 0.0   | -0.4  | 1.7   | 6.0   | 13.4  | 18.8  | 22.6  | 22.2  | 19.2  | 13.8  | 8.5   | 3.1  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 6.0   | 6.2   | 5.4   | 5.1   | 5.8   | 6.0   | 6.1   | 5.5   | 5.2   | 4.2   | 4.3   | 4.4  | (35) |
| (36) |                                                                                  |             | MCDBR | 9.4   | 8.5   | 7.6   | 9.3   | 10.7  | 9.8   | 8.6   | 7.8   | 7.4   | 6.5   | 7.2   | 6.9  | (36) |
| (37) |                                                                                  |             | MCWBR | 9.7   | 8.3   | 6.9   | 7.3   | 6.7   | 6.1   | 5.0   | 4.6   | 5.2   | 5.6   | 7.1   | 7.3  | (37) |
| (38) |                                                                                  | 5% WB       | MCDWR | 9.4   | 8.3   | 7.3   | 9.1   | 9.8   | 8.6   | 7.5   | 6.7   | 6.6   | 6.1   | 7.2   | 6.9  | (38) |
| (39) |                                                                                  |             | MCWBR | 9.7   | 8.3   | 6.9   | 7.8   | 6.9   | 6.1   | 4.8   | 4.5   | 5.2   | 5.7   | 7.3   | 7.3  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.294 | 0.287 | 0.319 | 0.349 | 0.367 | 0.382 | 0.411 | 0.391 | 0.359 | 0.338 | 0.312 | 0.294 | (40) |      |
| (41) |                                                                                  | taud        | 2.378 | 2.290 | 2.300 | 2.419 | 2.440 | 2.403 | 2.368 | 2.414 | 2.494 | 2.521 | 2.518 | 2.478 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 794   | 882   | 903   | 906   | 898   | 883   | 850   | 854   | 855   | 821   | 775   | 758   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 71    | 94    | 109   | 108   | 109   | 114   | 117   | 107   | 91    | 76    | 63    | 58    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.16  | 2.01  | 3.31  | 4.32  | 5.20  | 5.61  | 5.64  | 4.95  | 3.82  | 2.37  | 1.23  | 0.86  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.10  | 0.24  | 0.26  | 0.38  | 0.48  | 0.43  | 0.41  | 0.33  | 0.33  | 0.22  | 0.15  | 0.09  | (45) |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (27 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page